

**Examen**Durée: 2^h**Analyse II****Exercice I**

Calculer les primitives suivantes :

1) $\int \frac{x^2}{x^3 + 1} dx.$ ✖

2) $\int \cos(x) \sin(x) dx.$

3) $\int \sqrt{1-x} dx.$ ✖

4) $\int x\sqrt{1+x^2} dx.$

Exercice II

A l'aide d'une intégration par partie calculer les intégrales suivantes :

1) $I_1 = \int_1^e t \ln(t) dt$

2) $I_2 = \int_0^{\frac{\pi}{2}} t \sin(t) dt$

3) $I_3 = \int_0^{\sqrt{3}} 3t^2 \ln(t^2 + 1) dt$

Exercice III

1) Décomposer en éléments simple

$$f(x) = \frac{2x+1}{x^2(x+1)}.$$

2) Calculer

$$F(t) = \int_1^t \frac{1}{x^2} \ln(x^2 + x) dx. \quad \checkmark$$

Exercice IV

Calculer les intégrales suivantes en utilisant le changement de variable précisé :

1) $A = \int_1^4 \frac{1-\sqrt{t}}{t} dt$, en posant $u = \sqrt{t}$.

2) $B = \int_1^e \frac{(\ln(t))^n}{t} dt$, en posant $u = \ln(t)$.

3) $C = \int_0^{\frac{\pi}{2}} (\cos(t))^3 \sin(t) dt$, en posant $u = \sin(t)$.

Examen 2024 :



$$1) \int \frac{x^2}{x^3+1} dx$$

$$\frac{1}{3} \int \frac{3x^2}{x^3+1} dx$$

$$\frac{1}{3} \ln|x^3+1| + C$$

$$2) \int \cos x \sin x dx$$

$$\frac{1}{2} \sin^2(x) + C$$

$$3) \int \sqrt{1-x} dx$$

$$\int (1-x)^{\frac{1}{2}} dx$$

$$-\int (1-x)^{\frac{1}{2}} dx$$

$$-\frac{1}{\frac{1}{2}+1} (1-x)^{\frac{1}{2}+1} + C$$

$$-\frac{2}{3} (1-x)^{\frac{3}{2}} + C$$

$$4) \int x \sqrt{1+x^2} dx$$

$$\int x (1+x^2)^{\frac{1}{2}} dx$$

$$\frac{1}{2} \int 2x (1+x^2)^{\frac{1}{2}} dx$$

$$\frac{1}{2} \left(\frac{1}{\frac{1}{2}+1} (1+x^2)^{\frac{1}{2}+1} \right) + C$$

$$\frac{1}{2} \cdot \frac{2}{3} (1+x^2)^{\frac{3}{2}} + C$$

$$\frac{1}{3} (1+x^2)^{\frac{3}{2}} + C$$



$$I_1 = \int_1^e t \ln(t) dt$$

$$u = \ln t$$

$$u' = \frac{1}{t}$$

$$v' = t$$

$$v = \frac{1}{2} t^2$$

$$\left[\frac{1}{2} t^2 \ln t \right]_1^e - \int_1^e \frac{1}{t} \cdot \frac{1}{2} t^2 dt$$

$$\left[\frac{1}{2} t^2 \ln t \right]_1^e - \int_1^e \frac{1}{2} t dt$$

$$\left[\frac{1}{2} t^2 \ln t \right]_1^e - \frac{1}{2} \left[t^2 \right]_1^e$$

$$\left[\frac{1}{2} t^2 \ln t \right]_1^e - \frac{1}{2} \left[\frac{1}{2} t^2 \right]_1^e$$

$$\frac{1}{2} e^2 \ln e - \frac{1}{2} \ln 1 - \frac{1}{4} e^2 - \frac{1}{4}$$

$$\frac{1}{2} e^2 - \frac{1}{4} e^2 - \frac{1}{4} = \frac{1}{4} e^2 - \frac{1}{4}$$



$$I_2 = \int_0^{\frac{\pi}{2}} t \sin t \, dt$$

$$u' = 1$$

$$v' = \sin t$$

$$v = -\cos t$$

$$[-t \cos t]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\cos t \, dt$$

$$[-t \cos t]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos t \, dt$$

$$[-t \cos t]_0^{\frac{\pi}{2}} + [\sin t]_0^{\frac{\pi}{2}}$$

$$-\frac{\pi}{2} \cos \frac{\pi}{2} - 0 \cos 0 + \sin \frac{\pi}{2} - \sin 0$$

$$I_2 = 1$$

$$I_3 = \int_0^{\sqrt{3}} 3t^2 \ln(t^2+1) \, dt$$

$$u = \ln(t^2+1)$$

$$u' = \frac{2t}{t^2+1}$$

$$[t^3 \ln(t^2+1)]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} t^3 \cdot \frac{2t}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} \frac{2t^4}{t^2+1} \, dt = 2 \int_0^{\frac{\pi}{2}} \frac{t^2(t^2+1-1)}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} \frac{t^2(t^2+1) - t^2}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} \frac{t^2(t^2+1) - t^2}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} \frac{t^2(t^2+1) - t^2}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} t^2 - \frac{t^2}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} t^2 - \frac{t^2+1-1}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} t^2 - \frac{t^2+1}{t^2+1} - \frac{1}{t^2+1} \, dt$$

$$\int_0^{\frac{\pi}{2}} \left[t^3 \right]_0^{\frac{\pi}{2}} - [t]_0^{\frac{\pi}{2}} [\arctan t]_0^{\frac{\pi}{2}}$$

$$I_3 = [t^3 \ln(t^2+1)]_0^{\frac{\pi}{2}} - \left[\frac{t^3}{3} \right]_0^{\frac{\pi}{2}} + [t]_0^{\frac{\pi}{2}}$$

$$+ [\arctan]_0^{\frac{\pi}{2}}$$

$$u' = 3t^2$$

$$v = t^3$$



3 :

$$\frac{x+1}{x^2(x+1)}$$

$$\frac{x+1}{x^2(x+1)} = \frac{ax+b}{x^2} + \frac{c}{x+1}$$

$$\frac{x+1}{x^2(x+1)} = \frac{ax}{x^2} + \frac{b}{x^2} + \frac{c}{x+1}$$

$$\frac{x+1}{x^2(x+1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x+1}$$

on Multiplie par $(x+1)$

$$\frac{(x+1)(x+1)}{x^2(x+1)} = \frac{a(x+1)}{x} + \frac{b(x+1)}{x^2} + \frac{c(x+1)}{(x+1)}$$

$$\frac{x+1}{x^2} = \frac{a(x+1)}{x} + \frac{b(x+1)}{x^2} + c$$

Pour $x = -1$

$$-1 = c$$

$$\Rightarrow c = -1$$

on Multiplie par x^2

$$\frac{(x+1)x^2}{x^2(x+1)} = \frac{ax^2}{x} + \frac{bx^2}{x^2} + \frac{cx^2}{x+1}$$

$$\frac{x+1}{x+1} = ax + b + \frac{cx^2}{x+1}$$



Pour $x=0$

$$1 = b \Rightarrow b=1$$

$$\frac{2n+1}{x^2(x+1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x+1}$$

on Multiplie par x

$$\frac{(2x^2+1)x}{x^2(x+1)} = \frac{ax}{x} + \frac{bx}{x^2} + \frac{cx}{x+1}$$

$$\frac{2x+1}{x(x+1)} = a + \frac{b}{x} + \frac{cx}{x+1}$$

$$\lim_{x \rightarrow \infty} \frac{2x}{x^2} =$$

$$\lim_{x \rightarrow \infty} \frac{2}{x} =$$

$$0 = a + c$$

$$a + c = 0$$

$$a = -c$$

$$\underline{a=1}$$

$$\frac{2n+1}{x^2(x+1)} = \frac{1}{x} + \frac{1}{x^2} - \frac{1}{x+1}$$



Solution:

$$I(t) = \int_2^t \frac{1}{x^2} \ln(x^2+x) dx$$

$$u = \ln(x^2+x) \quad u' = \frac{1}{x^2}$$

$$u' = \frac{2x+1}{x^2+x} \quad u = -\frac{1}{x}$$

$$\left[-\frac{1}{x} \ln(x^2+x) \right]_2^t + \int_2^t \frac{1}{x} \frac{2x+1}{x^2+x} dx$$

$$\left[-\frac{1}{x} \ln(x^2+x) \right]_2^t + \int_2^t \frac{2x+1}{x^2(x+1)} dx$$

$$\int_2^t \frac{2x+1}{x^2(x+1)} dx$$

$$= \int_2^t \frac{1}{x} + \frac{1}{x^2} - \frac{1}{x+1} dx$$

$$[\ln x]_2^t + \left[-\frac{1}{x} \right]_2^t - [\ln|x+1|]_2^t$$

$$\left[-\frac{1}{x} \ln(x^2+x) \right]_2^t + [\ln x]_2^t + \left[-\frac{1}{x} \right]_2^t$$

$$- [\ln|x+1|]_2^t$$

$$-\frac{1}{t} \ln(t^2+t) - \ln 2 + \ln t - \frac{1}{t} + \frac{1}{2}$$

$$- \ln|t+1| + \ln|2|$$



$$u = \sqrt{t}$$

$$\text{Si } t=1 \rightarrow u=1$$

$$\text{Si } t=4 \rightarrow u=2$$

$$u = \sqrt{t}$$

$$du = \frac{1}{2\sqrt{t}} dt$$

$$dt = 2\sqrt{t} du$$

$$\int_1^2 \frac{1-u}{u^2} 2u du$$

$$2 \int_1^2 \frac{1-u}{u} du$$

$$2 \int_1^2 \frac{1}{u} - 2 \int_1^2 1 du$$

$$2 [\ln u]_1^2 - 2 [u]_1^2$$

$$2 \ln 2 - 4 + 2$$

$$2 \ln 2 - 2$$

$$a) \int_1^e \frac{(\ln t)^n}{t} dt$$

$$u = \ln t$$

$$\text{Si } t=1 \quad u=0$$

$$\text{Si } t=e \quad u=1$$

$$u = \ln t$$

$$du = \frac{1}{t} dt$$

$$dt = t du$$

$$\int_0^1 \frac{u^n}{t} t du$$

$$\int_0^1 \frac{1}{n+1} u^{n+1} du$$

$$\frac{1}{n+1} \left[\frac{u^{n+1}}{n+1} \right]_0^1$$

$$3) \int_{\frac{\pi}{2}}^{\pi} (\cos t)^3 \sin t dt$$

$$u = \sin t$$

$$du = \cos t dt$$

$$dt = \frac{du}{\cos t}$$

$$\text{Si } t = \frac{\pi}{2} \quad u=1$$

$$\text{Si } t = \pi \quad u=0$$

$$\int_1^0 \cancel{\cos t} \cos^2 t u \frac{du}{\cancel{\cos t}}$$



$\sin t = u$

$$\sin^2 t + \cos^2 t = 1$$

$$\cos^2 t = 1 - \sin^2 t \\ = 1 - u^2$$

$$\int_0^{\pi/2} (1 - u^2) u \, du$$

$$\int_0^{\pi/2} u - u^3 \, du$$

$$\left[\frac{1}{2} u^2 - \frac{u^4}{4} \right]_0^{\pi/2}$$

$$\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$\frac{1}{4}$

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